

CoBit on Sudoku: sampling study and best-achievable accuracy

Continuous bitstream diffusion (CoBit) under the S-FLM verifiable-task protocol. What sampler, churn, and NFE settings maximise exact-match — and how far past the published headline they reach. 20 June 2026

Task and metric. We replicate the Sudoku benchmark of *S-FLM* (Deschenaux & Gulcehre). Each puzzle is a 9×9 grid serialised to 180 tokens (12-symbol vocab, 4 bits/token $\rightarrow$ a 720-bit bitstream); the 91-token prompt (clues) is the clean conditioning prefix, clamped at every solver step and excluded from the loss. The metric is *exact match*: the entire 89-token solution suffix must be byte-for-byte correct. Difficulty = #clues: easy/medium/hard = 40/35/30. Data split, decoding, and metric are identical to S-FLM, so the numbers are directly comparable to their reported results.

Method under test. CoBit is a variance-exploding continuous diffusion over the raw binary bitstream ($\sigma \in [0.002, 80]$, $\rho=7$, EDM preconditioning $c_{\text{in}} = 1/\sqrt{\sigma^2 + \sigma_{\text{data}}^2}$). The denoiser is an 8-block/512-d/ 8-head transformer matched to S-FLM. All evaluation uses the *raw* (running) training weights: at the 20k-step budget the EMA decay 0.9999 has not converged off the noisy early average, so EMA decoding falls out-of-vocabulary — expected at this step count, not a defect.

The samplers. All sit in one reverse-SDE family, $dx = (1 + \lambda(\sigma)) \sigma s_\theta dr + \sqrt{2\lambda(\sigma)\sigma} dW_r$, discretised on the model's *entropy-rate* σ -grid (sampling points placed uniformly in the entropy-rate CDF, concentrating steps where bits resolve):

- **DDIM (deterministic)** — probability-flow ODE, $\lambda=0$. One denoiser call/step.
- **DDIM + EDM churn** — adds noise implicitly via $\hat{\sigma} = (1+\gamma)\sigma$; effective Langevin strength $\lambda_i \approx \gamma\sigma_i/\Delta_i$, gated by the entropy-rate density. *Hard-capped* at $\gamma \leq \sqrt{2} - 1 \approx 0.414$. The headline path.
- **Euler–Maruyama (EM) / Predictor–Corrector (PC)** — explicit entropy-gated SDE samplers (this work). Noise is injected in x -space at the on-grid σ via a λ_0 -scaled entropy-rate profile; *no churn cap*. At $\lambda_0=0$ they reduce to deterministic DDIM bit-identically. PC splits each step into a PF predictor + a Langevin corrector. Classifier-free guidance, when used, is applied *predictor-only* (corrector runs at the conditional score), which avoids the $(1+\lambda)$ over-amplification that naive guide-everywhere CFG suffers under stochastic sampling.

Stochasticity is essential. On the entropy-rate grid, EDM churn turns a constant churn budget into Langevin correction concentrated at the high-information noise band where a premature deterministic bit-decision would otherwise lock in an error. Sweeping γ at NFE=180 (mean over seeds 42/43/44) lifts hard from 40.2% ($\gamma=0$, prob.-flow) to a plateau $\sim 65.9\%$ near $\gamma \approx 0.4$ — **+25.7 pts from churn alone**. Easy/medium gain +5.2/ +15.0. Churn is the single most important sampling choice on this task.

Headline at NFE=180 vs. diffusion-LM baselines. At the standard 180-NFE operating point (best γ , EDM churn, entropy grid, mean over 3 seeds), CoBit **beats every published DLM baseline on all three difficulties**:

Method (180 NFE)	easy	medium	hard
CoBit (best γ)	98.5	91.7	65.9
Duo	96.3	84.7	58.4
S-FLM (+trunc+adaptive)	94.8	85.2	45.0
FLM	94.2	82.7	44.5
AR Transformer (greedy)	14.6	5.1	1.0

Better noise shape: EM/PC vs. EDM churn. At *matched* stochasticity ($\lambda_0=70$, the entropy-rate-*as_saved* anchor $\approx$ EDM $\gamma=0.4$), the explicit entropy-gated PC sampler equals or beats EDM churn at NFE=180 (single seed, $n=2000$, raw): easy 98.20 vs 97.90; **hard 64.55 vs 61.40 (+3.2)**. The reverse-SDE Langevin is a better-shaped correction than implicit churn, most visibly on hard. (Guidance itself did not improve Sudoku — it is a near-saturated conditional task — but the predictor-only formulation keeps it *harmless*, whereas naive guide-everywhere CFG degraded both exact-match and validity as the weight grew.)

Scaling NFE — the largest lever, especially on hard. Holding $\gamma=0.4$ (so the cumulative churn $S_{\text{churn}} = \gamma(\text{NFE}-1)$ scales up with NFE) and increasing the number of solver steps yields monotone gains that are largest where the deterministic floor is lowest (single seed, $n=2000$, raw, EDM churn, entropy grid):

exact-match %	NFE 180	384	768	1024	Δ best
easy	97.90	98.00	98.75	98.60	+0.85
medium	92.85	93.85	94.75	94.95	+2.10
hard	61.40	64.75	67.50	67.40	+6.10

Hard improves +6.1 pts (valid-sudoku rate +8.9), saturating by $\text{NFE} \approx 768$; medium is still creeping up at 1024; easy peaks at 768. The plateau on hard sits at $\gamma=0.4$ — i.e. *near the EDM churn cap* — which indicates hard is now limited by the churn ceiling, not by step count: the motivation for the uncapped EM sampler.

Best achievable here. Combining the better-regularised checkpoint (trained with 0.1 conditioning dropout), the entropy-grid stochastic sampler, and higher NFE pushes past our own 180-NFE headline and well past the prior DLM frontier:

	easy	medium	hard
Best prior DLM (180 NFE)	96.3	85.2	58.4
CoBit headline (180 NFE, 3-seed)	98.5	91.7	65.9
CoBit best sampling (this work)	98.75	94.95	67.50
best setting	NFE768	NFE1024	NFE768

Caveats. The “best sampling” row is single-seed (42) on the conditioning-dropout checkpoint with γ fixed at 0.4 (not re-optimised per NFE); seed variance at 180 NFE was tight ($\leq 1.3\%$) and γ was not tuned upward, so these are a conservative lower bound on the achievable accuracy. EM/PC at $\lambda_0=0$ reproduce deterministic DDIM bit-identically (validation gate), confirming the new samplers are correct.

Conclusions — what sampling settings work on Sudoku.

1. **Use stochastic sampling.** EDM churn (or the EM/PC reverse-SDE) on the entropy-rate grid is decisive: +26 pts on hard over deterministic probability-flow. Deterministic sampling is never the right choice here.
2. **Push churn to the high end.** Accuracy rises monotonically with γ and plateaus near $\gamma \approx 0.4$ – 0.5 ; on hard the plateau coincides with the EDM cap $\sqrt{2} - 1$, so the uncapped EM sampler is the way to test still-higher effective churn.
3. **Scale NFE, prioritising hard.** More solver steps (with churn scaled accordingly) give the largest single gain on the hardest puzzles (+6 pts to $\text{NFE} \approx 768$); easy/medium saturate earlier. NFE is a genuine accuracy axis, not just a cost axis, for low-baseline difficulties.
4. **Prefer the entropy-gated SDE (EM/PC) over implicit EDM churn** at matched stochasticity — it is at least as good and better on hard, and (unlike EDM) is uncapped, enabling the high-effective-churn $\times$ high-NFE regime on the hardest problems.
5. **Use raw weights** at this training budget, and if guidance is ever applied, use the *predictor-only* formulation (guidance does not help Sudoku but predictor-only keeps it safe).